Gift of the Givers · Relief Management System

DELIVERABLE 05

Backend Schema

Database design, constraints, relationships, derived values and migration history for SQL Server / Azure SQL.

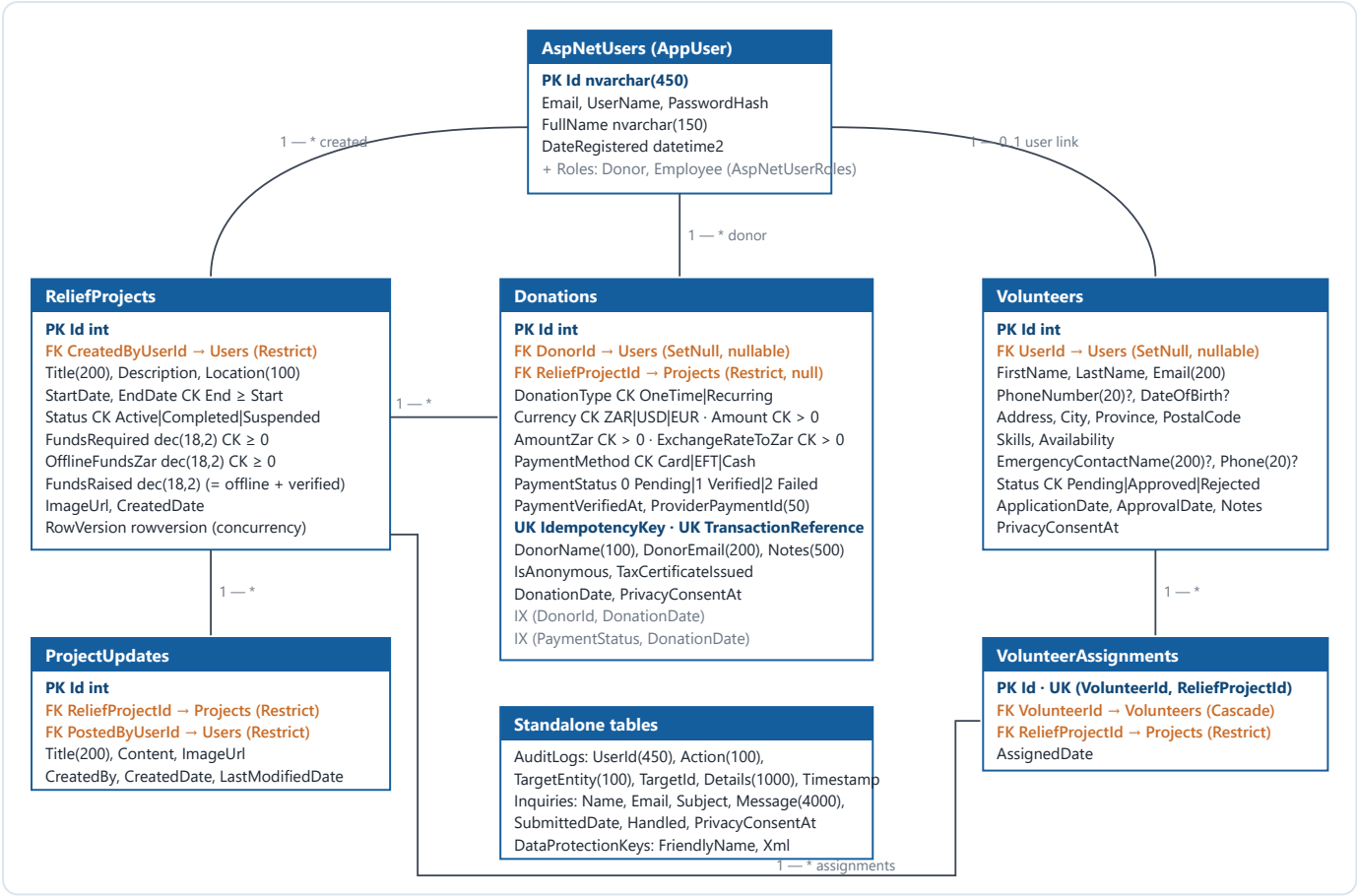
DOCUMENT	VERSION	DATE	STATUS
D-05	2.0	2026-11-02	Current — at migration AuditRemediationNov2026

CONTENTS

1. Overview
2. Tables
3. Relationships and delete behaviour
4. Derived values and invariants
5. Concurrency
6. Migration history
7. Data lifecycle and privacy
8. Seed data (Development only)

1. Overview

- **Engine:** SQL Server 2022, Azure SQL, or LocalDB, through EF Core 10. **Migrations are the source of truth**; CI fails if the model and migrations differ.
- **Context:** `GiftOfTheGivers/Data/ApplicationDbContext.cs`, which extends `IdentityDbContext<AppUser>` and implements `IDataProtectionKeyContext`.
- **Integrity strategy:** constraints in the database (check, unique, foreign key) plus validation in the application. Money is `decimal`; time is UTC `datetime2`; concurrency uses `rowversion`.



Entity relationship diagram

2. Tables

2.1 ReliefProjects

Column	Type	Null	Rules
Id	int identity	No	PK
CreatedByUserId	nvarchar(450)	Yes	FK → AspNetUsers (Restrict)
Title	nvarchar(200)	No	
Description	nvarchar(max)	No	Input limited to 2,000 characters
Location	nvarchar(100)	No	
StartDate / EndDate	datetime2	No / Yes	`CK_ReliefProject_Dates` : EndDate is null or ≥ StartDate
Status	nvarchar(max) (enum as string)	No	`CK_ReliefProject_Status` : Active, Completed, Suspended
FundsRequired	decimal(18,2)	No	`CK_ReliefProject_FundsRequired_NonNegative`
OfflineFundsZar	decimal(18,2)	No	`CK_ReliefProject_OfflineFunds_NonNegative` ; staff-maintained
FundsRaised	decimal(18,2)	No	**Derived**: OfflineFundsZar + Σ AmountZar of Verified donations
ImageUrl	nvarchar(max)	Yes	http(s) only, ≤ 500 characters (validated)
RowVersion	rowversion	—	Optimistic concurrency token
CreatedDate	datetime2	No	UTC

2.2 Donations

Column	Type	Null	Rules
Id	int identity	No	PK (internal only, never in public URLs)
DonorId	nvarchar(450)	Yes	FK → AspNetUsers (**SetNull** when the user is deleted)
ReliefProjectId	int	Yes	FK → ReliefProjects (Restrict); null = General Fund
DonationType	nvarchar(20)	No	`CK_Donation_Type` : OneTime, Recurring (pledge)
Currency	nvarchar(3)	No	`CK_Donation_Currency` : ZAR, USD, EUR
Amount	decimal(18,2)	No	`CK_Donation_Amount` > 0; in the chosen currency
AmountZar	decimal(18,2)	No	`CK_Donation_AmountZar` > 0; charged to PayFast; used for all totals
ExchangeRateToZar	decimal(18,6)	No	`CK_Donation_ExchangeRate` > 0; 1 for ZAR
PaymentMethod	nvarchar(50)	No	`CK_Donation_PaymentMethod` : Card, EFT, Cash
PaymentStatus	int	No	0 Pending, 1 Verified, 2 Failed
PaymentVerifiedAt	datetime2	Yes	Set by a verified ITN

Column	Type	Null	Rules
ProviderPaymentId	nvarchar(50)	Yes	PayFast `pf_payment_id`
IdempotencyKey	uniqueidentifier	No	**Unique**; one donation per form submission
TransactionReference	nvarchar(100)	Yes	**Unique**; 32 hex characters; PayFast `m_payment_id` ; public receipt key
DonorName / DonorEmail	nvarchar(100) / nvarchar(200)	Yes	
IsAnonymous	bit	No	Hides the name on certificates and reports
TaxCertificateIssued	bit	No	True once verified
Notes	nvarchar(500)	Yes	
DonationDate	datetime2	No	UTC
PrivacyConsentAt	datetime2	Yes	Null only for rows created before remediation

Indexes: `DonorId` , `ReliefProjectId` , `(DonorId, DonationDate)` , `(PaymentStatus, DonationDate)` , and unique `IdempotencyKey` and `TransactionReference` .

2.3 Volunteers

Column	Type	Null	Rules
Id	int identity	No	PK
UserId	nvarchar(450)	Yes	FK → AspNetUsers (SetNull)
FirstName / LastName	nvarchar(100)	No	
Email	nvarchar(200)	No	Indexed
PhoneNumber	nvarchar(20)	**Yes**	Optional, never invented
Address / City / Province / PostalCode	200 / 100 / 50 / 10	Yes	
DateOfBirth	datetime2	**Yes**	Optional; ≥ 16 years if given
Skills / Availability	nvarchar(max)	No	Input limited to 2,000 / 200 characters
EmergencyContactName / Phone	nvarchar(200) / nvarchar(20)	Yes	
Status	nvarchar(450) (enum as string)	No	`CK_Volunteer_Status` : Pending, Approved, Rejected; index `(Status, ApplicationDate)`
ApplicationDate / ApprovalDate	datetime2	No / Yes	UTC
Notes	nvarchar(max)	Yes	
PrivacyConsentAt	datetime2	Yes	

2.4 Other tables

Table	Key columns	Notes
VolunteerAssignments	VolunteerId (Cascade), ReliefProjectId (Restrict), AssignedDate	Unique `(VolunteerId, ReliefProjectId)`
ProjectUpdates	ReliefProjectId (Restrict), PostedByUserId (Restrict), Title(200), Content, ImageUrl, CreatedBy, CreatedDate, LastModifiedDate	
Inquiries	Name(200), Email(200), Subject(200), Message(4000), SubmittedDate, Handled, PrivacyConsentAt	
AuditLogs	UserId(450), Action(100), TargetEntity(100), TargetId, Details(1000), Timestamp	Index `(TargetEntity, TargetId)` and `Timestamp`. Payment events use actor `system:payfast`
DataProtectionKeys	FriendlyName, Xml	ASP.NET Core key ring shared by all instances
AspNet*	Identity schema (key length 450)	`Stores.MaxLengthForKeys = 0` keeps the runtime model matching this schema

3. Relationships and delete behaviour

Relationship	Cardinality	On delete	Reason
User → Donations	1 : * (optional)	SetNull	Deleting an account must not delete financial records
User → Volunteers	1 : 0..1 (optional)	SetNull	Same
User → Projects (creator), Updates (poster)	1 : *	Restrict	Keep authorship; deactivate staff instead of deleting
Project → Donations	1 : *	Restrict	Financial history must survive
Project → Updates, Assignments	1 : *	Restrict	Avoid multiple cascade paths in SQL Server
Volunteer → Assignments	1 : *	Cascade	Assignments have no meaning without the volunteer

4. Derived values and invariants

Invariant	Enforced by
`FundsRaised = OfflineFundsZar + Σ(AmountZar where PaymentStatus = Verified)`	`DonationService` (verification), offline-funds delta on edit, `RecalculateFundsRaisedAsync`, Funding Reconciliation page
A donation is counted once	Status check (Verified short-circuits) + rowversion retry
One donation per form submission	Unique `IdempotencyKey`, plus catch-and-return of the winning row
Certificate only for paid gifts	Query filter `PaymentStatus = Verified` in every certificate path
Staff changes are traceable	`AuditService.Record` staged in the same `SaveChanges`

5. Concurrency

`ReliefProjects.RowVersion` protects totals. Payment verification and recalculation retry up to three times on `DbUpdateConcurrencyException`. A project edit that conflicts shows "changed by someone else, reload". The EF SQL Server retry strategy (`EnableRetryOnFailure`) handles transient faults. No user-initiated transactions are used, so it is compatible.

6. Migration history

Migration	Summary	Notes
20260828181123_InitialCreate	Identity, projects, donations, volunteers, updates, assignments	
20260924181920_SyncCurrentModel	RowVersion, PaymentStatus, FK changes	
20260924191049_P1PaymentsEmailAndStatuses	DonorEmail, IdempotencyKey, AuditLogs, Inquiries, check constraints	**Edited during remediation (F-21):** the `AlterColumn` on `RowVersion` was removed because SQL Server cannot alter a rowversion column, so the migration failed on every empty database
20260924195240_P0Phase0CriticalFixes	Donor FK; raw-SQL check constraints	The raw constraints duplicated EF-managed ones (dropped in the remediation migration)
20260924200101_FixPendingModelChanges	Empty	Symptom of drift-chasing under the suppressed warning
20260925180656_SyncDonorCascade	Donor FK SetNull	
(removed) 20260924190000_P0SecurityAndDonationIntegrity	—	Orphan file without a `[Migration]` attribute; never discovered by EF; deleted
20261102112731_AuditRemediationNov2026	AmountZar, ExchangeRateToZar, DonorName, PaymentVerifiedAt, ProviderPaymentId, PrivacyConsentAt (×3), OfflineFundsZar, nullable DOB/phone, sized audit columns + Details, DataProtectionKeys, new check constraints and index	Backfills rand values (ZAR exact; USD 18.50 and EUR 20.00 indicative), normalises PaymentMethod (`PayPal` → `Card`), sets PaymentVerifiedAt for verified rows, drops the duplicate constraints. Down re-creates them

Verification (2026-11-02, LocalDB): old chain applied to an empty database, legacy rows inserted (ZAR, USD with PayPal, EUR), upgrade applied, values checked, rolled back and re-applied, drift check clean. CI repeats the empty-database apply plus rollback/re-apply on SQL Server 2022 for every push.

7. Data lifecycle and privacy

Data	Retention (proposed, pending legal review)	Deletion path
Donations	5 years after the end of the tax year	Not deleted; anonymise donor fields on request after retention
Volunteers	2 years after last activity	Staff deletion (to be built, T-2.4)
Inquiries	1 year after handled	Scheduled purge (to be built, T-2.4)
Accounts	Until the user deletes them	Identity *Delete personal data*; donations keep SetNull links
Audit logs	5 years	Append-only

8. Seed data (Development only)

Roles `Donor` and `Employee` are created in every environment. In Development (or with `Seed:DemoData=true`): two demo accounts, six projects with offline funds, three volunteers (two approved, assigned to two projects), and three donations (two verified, one pending) so dashboards and reconciliation show both states.